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| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 06/02/2009 |
| **PI Name:** | Hoel, David G Ph.D. | | |
| **Project Title:** | The Analysis of the Potential Health Risks from High LET Radiation | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Charleston | **State:** | SC |
| **Zip Code:** | 29425-2203 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 02-OBPR-02 |
| **Start Date:** | 05/19/2003 | **End Date:** | 05/31/2008 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 5/31/2008 per P. Denkins/JSC (5/2007)<br>NOTE: NCE to 5/19/2007 per J. Dardano (4/06) | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Carnes, Bruce ( University of Oklahoma ) | | |
| **Grant/Contract No.:** | NAG9-1518 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Extensive pathology and mortality data from Argonne National Laboratory for B6CF1 mice receiving single and protracted exposure to neutrons will be used to estimate radiation-induced risk (mortality and morbidity) for heavy ions. The justifications for using neutron data as a surrogate for HZE particles are that the 0.85 MeV JANUS neutrons are a high LET radiation, and their biological effectiveness and dose-response at lower doses have been shown to be similar to those for Fe and Argon. Although we will acquire heavy ion mortality data as it becomes available, the neutron data will be used to fit both empirical and biologically motivated dose-response models. Empirical models will be examined because they have already been applied to these data in order to successfully make interspecies predictions (mouse to dog to human) of radiation-induced mortality for low LET radiation (both acute and protracted). The biological models will be examined to not only compare their predictions to the empirical models, but also to examine how knowledge of the pathogenesis of disease processes affects quantitative models used to predict radiation-induced mortality risks. Finally, the results of this modeling project will be synthesized into practical estimates of the potential radiation risks for astronauts on prolonged space flights.<br><br>To obtain neutron RBE values for a variety of cancer and cardiovascular endpoints. Evaluate the assumption of a constant RBE over dose levels. Examine the predictability of gamma exposed animals to human cancer risk.<br><br>Will provide needed information for estimating astronaut cancer risks from HZE exposures in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Help to set exposure limits for workers exposed to high LET radiation. |
| **Task Progress:** | Of concern to NASA is the possible non-cancer effects of radiation exposure such as heart disease. After long latency period increased mortality from heart disease has been observed in the A-bomb survivors. Very detailed cardiovascular pathology was carried out for one of the major Janus experiments involving both gamma and neutron exposures. Our analysis suggests that there is a neutron effect greater than what is observed with gamma exposures. Also it appears that low doses of neutron can affect cardiovascular disease (CVD) mortality while it is not clear that low-dose exposures of gamma in either the mouse or in humans is a CVD risk.<br>The biologically based two-stage clonal expansion (TSCE) model needed changes before comparisons are appropriate between gamma and neutron as well as humans. The current model assumes that the mutation rates are permanently changed by the exposure. This certainly cannot be the case for mutations other than the initial one affecting normal stem cells. The model extensions were completed and the mouse data was analyzed. The neutron effects compared to gamma were similar to what has been previously believed to be the case. Using the non-generalized TSCE models Heidenreich et al 2006 showed that for lung cancer the neutron RBE was about 10. We notice that the data for the endpoints lung, lymphoreticular and solid tumors, the RBE was below 12 in contrast to what has been suggested by Kellerer et al 2006. It should however be noted that there was little low dose data for the gamma exposures in the mice and thus and gamma dose response curvature could substantially change the RBE estimates.<br><br>We did consider experimental data relating alpha to beta effects in the mouse (Priest et al 2006, 2008). By controlling internal exposures through the use of FAP particle we observed that the effect of alpha was not as great as is currently assumed.<br><br>[NOTE: Above added by Task Book editor from final report; 6/2009] |
| **Bibliography Type:** | Description: (Last Updated: 02/25/2014) |
| **Articles in Peer-reviewed Journals** | Hoel DG. "Ionizing radiation and cardiovascular disease." Ann N Y Acad Sci. 2006 Sep;1076:309-17. PMID: 17119211 , Sep-2006 |
| **Articles in Peer-reviewed Journals** | Priest ND, Hoel DG, Brooks PN. "Relative toxicity of chronic irradiation by 45Ca beta particles and 242Cm alpha particles with respect to the production of lung tumors in CBA/Ca mice." Radiat Res. 2006 Nov;166(5):782-93. PMID: 17067209 , Nov-2006 |
| **Articles in Peer-reviewed Journals** | Heidenreich WF, Carnes BA, Paretzke HG. "Lung cancer risk in mice: analysis of fractionation effects and neutron RBE with a biologically motivated model." Radiat Res. 2006 Nov;166(5):794-801. PMID: 17067205 , Nov-2006 |
| **Articles in Peer-reviewed Journals** | Priest ND, Hoel DG, Brooks PN. "Relative toxicity of (45)Ca beta-particles and (242)Cm alpha-particles following their intravenous injection into mice as radiolabelled FAP." Int J Radiat Biol. 2010 Apr;86(4):300-20. PMID: 20353340 , Apr-2010 |
| **Articles in Peer-reviewed Journals** | Hoel DG, Grahn D, Carnes BA. "Cardiovascular effects of fission neutron or 60Co gamma-ray exposure in the B6CF1 mouse." under revision, 2008. , Dec-2008 |
| **Articles in Peer-reviewed Journals** | Priest ND, Hoel D. "Determining the relative toxicity and RBE of internal emitters in animals." Health Phys. 2010 Sep;99(3):367-70. PMID: 20699699 , Sep-2010 |